

U.S. SEMICONDUCTOR: TRADE & LABOR ANALYSIS

U.S. Employment & Wages — Quarterly Census of Employment and Wages (BLS QCEW) | 2025

Overview

This document describes the data cleaning steps applied to the Bureau of Labor Statistics Quarterly Census of Employment and Wages (QCEW) dataset. The objective was to simplify the dataset so that it focuses on variables relevant to employment, wages, and contributions across industries and regions — removing redundant metadata, disclosure codes, and technical identifiers that added no analytical value, while retaining establishment counts, employment levels, wage measures, and over-the-year changes.

The cleaned output is saved as Cleaned Quarterly Census of Employment and Wages.csv and is ready for labor market and wage trend analysis.

Dataset Summary

Attribute	Value
Source file	Quarterly Census of Employment and Wages.csv
Output file	Cleaned Quarterly Census of Employment and Wages.csv
Data provider	U.S. Bureau of Labor Statistics (BLS)
Coverage	Employment, wages, and contributions by industry, ownership, and area
Duplicate rows	0 (confirmed)
Analytical focus	Establishments, employment levels, wage totals, contributions, average weekly wages, over-the-year changes

Step 1 — Reading the Raw Dataset

Objective

Load the original QCEW dataset into a pandas DataFrame for inspection and transformation.

Code

```
df = pd.read_csv('Quarterly Census of Employment and Wages.csv')
```

Key Design Choice

- **Raw file scope:** The file contained many fields, including aggregation codes, disclosure flags, and industry identifiers that were not essential for the intended labor market analysis.

Step 2 — Inspecting the Data

Objective

Confirm the dataset structure and ensure it loaded correctly before performing any transformations.

Code

```
print(df.head())
```

Key Design Choice

- **Head inspection:** Printing the first few rows provides an immediate visual check of column names, data types, and representative values — catching any loading issues (encoding errors, misaligned columns) before transformation begins.

Step 3 — Dropping Redundant or Misleading Columns

Objective

Remove columns that add no analytical value — specifically disclosure codes, aggregation metadata, and redundant classification identifiers.

Code

```
cols_to_drop = [
    'agglvl_code',
    'size_code',
    'disclosure_code',
    'lq_disclosure_code',
    'oty_disclosure_code',
    'industry_code'
]
df = df.drop(columns=cols_to_drop)
```

Rationale for Each Dropped Column

Column	Reason for Removal
disclosure_code	Suppression metadata — indicates data withholding rules, not analytical content
lq_disclosure_code	Location quotient disclosure flag — suppression metadata only
oty_disclosure_code	Over-the-year change disclosure flag — suppression metadata only
agglvl_code	Technical aggregation level identifier — not needed when working with the full dataset
size_code	Establishment size code — metadata without analytical value for wage/employment analysis
industry_code	Redundant — ownership and area codes already provide the necessary classification context

Step 4 — Checking Data Types

Objective

Verify that numeric fields are correctly recognized by pandas for calculations and aggregations.

Code

```
print(df.dtypes)
```

Key Design Choice

- **Type verification:** Ensuring that employment counts, wage totals, and change metrics are stored as numeric types (int or float) — rather than object/string — is essential for correct aggregation, averaging, and statistical analysis.

Step 5 — Checking for Duplicates

Objective

Confirm that no duplicate rows exist in the dataset to ensure accuracy in all aggregations and prevent inflated totals.

Code

```
print(df.duplicated().sum())
```

Result

Check	Result
Duplicate rows found	0
Action taken	None required — dataset is clean

Step 6 — Saving the Cleaned Dataset

Objective

Export the cleaned dataset as a structured CSV file ready for labor market analysis, wage trend modeling, and regional comparisons.

Code

```
df.to_csv('Cleaned Quarterly Census of Employment and Wages.csv', index=False)
```

Fields Retained in the Cleaned Dataset

Field Category	Description
Establishment counts	Number of reporting units by area, industry, and ownership
Employment levels	Monthly employment figures for the reference quarter
Wage totals	Total wages paid in the quarter
Average weekly wages	Derived wage metric for cross-industry and cross-region comparison
Location quotients	Industry concentration relative to national average

Field Category	Description
Over-the-year changes	Year-on-year change metrics for employment, wages, and establishments
Classification fields	Ownership type, area name, industry title, year, quarter

Key Decisions and Rationale

- **Disclosure codes dropped:** These fields were metadata for data suppression rules and added no analytical value to labor market or wage analysis.
- **Aggregation and size codes removed:** Technical identifiers that introduced unnecessary complexity without contributing to employment or wage insights.
- **Industry code excluded:** Since ownership and area codes already provide industry classification context, this field was redundant.
- **Duplicates confirmed absent:** Verified that no duplicate rows existed, ensuring reliable totals and averages across all aggregations.
- **Focus sharpened:** By retaining establishment counts, employment levels, wages, contributions, and change metrics, the dataset directly supports labor market and wage trend analysis.
- **Transparency ensured:** Every cleaning step is documented, making the dataset fully reproducible and audit-ready.

Data Source

[U.S. Bureau of Labor Statistics — Quarterly Census of Employment and Wages \(QCEW\)](#)